Changepoints for Environmental Data

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Plan

- What are changepoints?
- How to find changepoints?
- How many changes?
- Important Considerations

Workshop is practical side of what I present here.

What are Changepoints?

Changepoints are also known as:

- breakpoints
- segmentation
- structural breaks
- regime switching
- detecting disorder

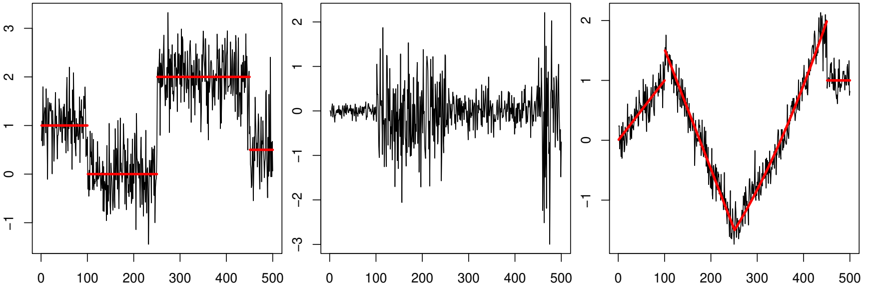
and can be found in a wide range of literature including

- quality control
- economics
- medicine
- environment
- linguistics
- ...

What are changepoints?

For data $y_1, \ldots, y_n$, if a changepoint exists at τ, then $y_1, \ldots, y_\tau$ differ from $y_{\tau+1}, \ldots, y_n$ in some way.

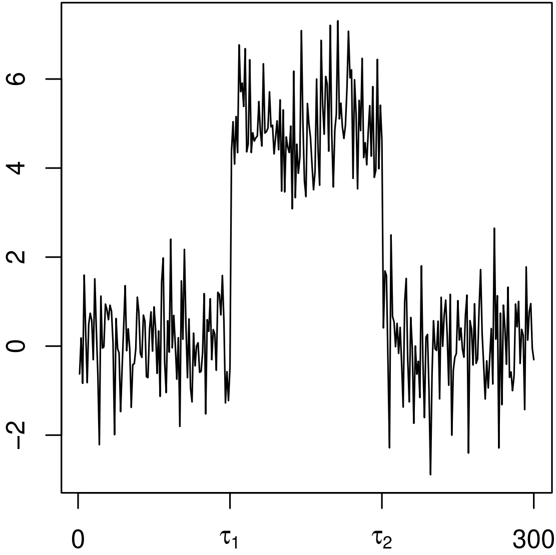
There are many different types of change.



What is the goal?

- Has a change occurred?
- If yes, where is the change?
- What is the difference between the pre and post change data?
  - Maybe this is the type of change
  - Maybe it is the parameter values before and after the change
- What is the probability that a change has occured?
- How certain are we of the changepoint location?
- How many changes have occurred (+ all the above for each change)?
- Why has there been a change?

Notation and Concepts



Notation and Concepts

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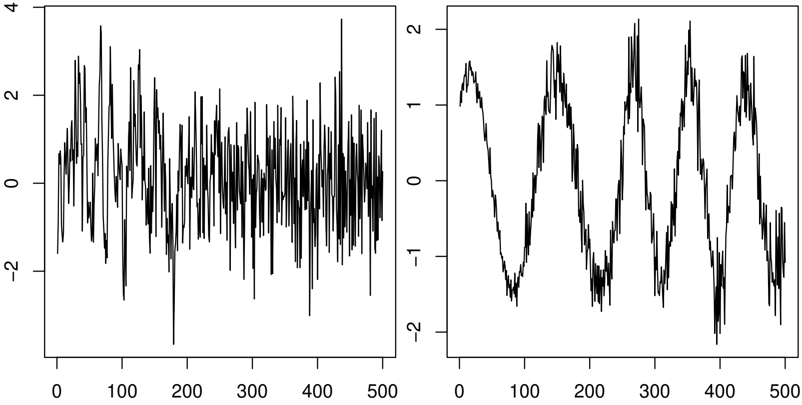
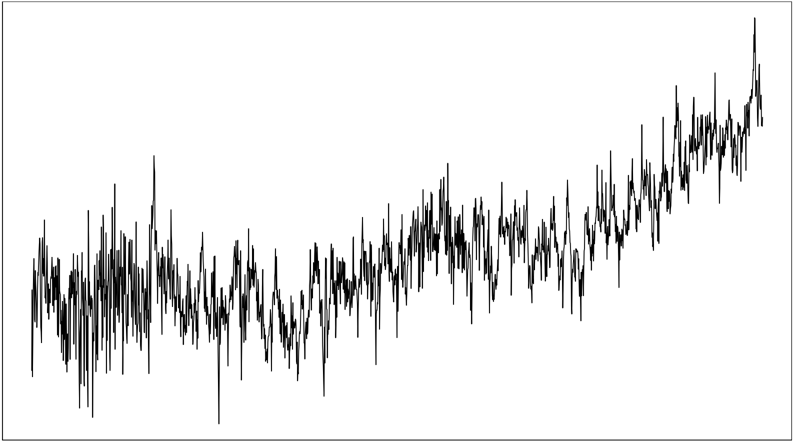
Thus a changepoint model for a change in mean has the following formulation:

$$y_t = \begin{cases} \mu_1 & \text{if} \quad 1 \leq t \leq \tau_1 \\ \mu_2 & \text{if} \quad \tau_1 < t \leq \tau_2 \\ \vdots & \quad\quad \vdots \\ \mu_{m+1} & \text{if} \quad \tau_m < t \leq \tau_{m+1} = n \end{cases}$$

More complicated changes



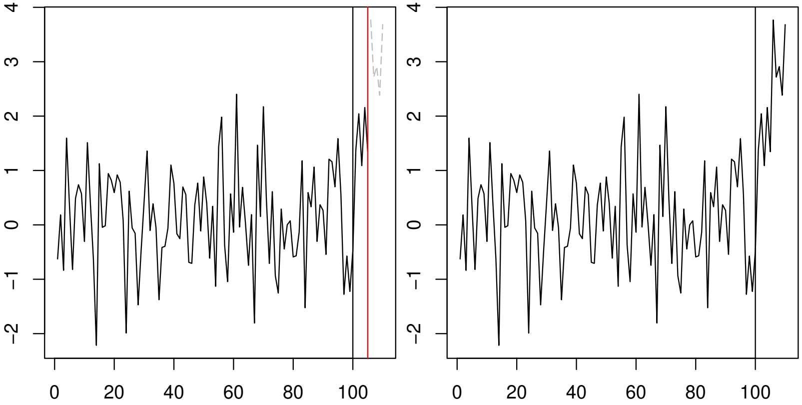
More like reality



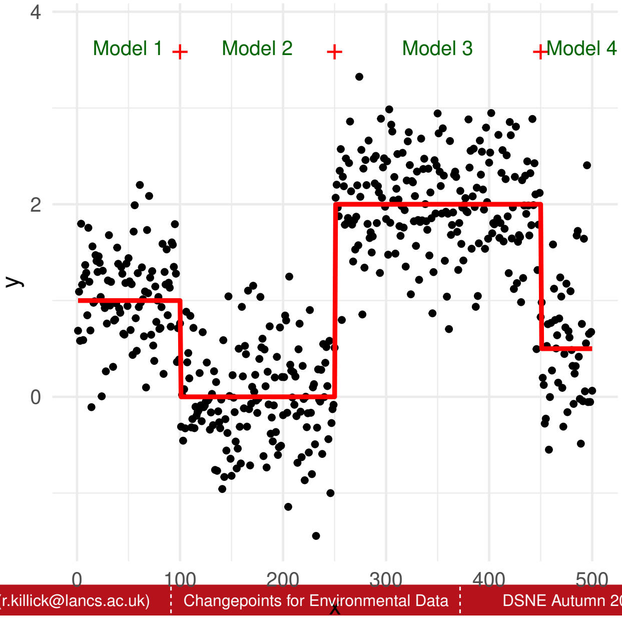
Online vs Offline

- Online
  - Processes data as it arrives or in batches
  - Goal is quickest detection of a change
  - Often used in processing control, intrusion detection
- Offline
  - Processes all the data in one go
  - Goal is accurate detection of a change
  - Often used in genome analysis, audiology

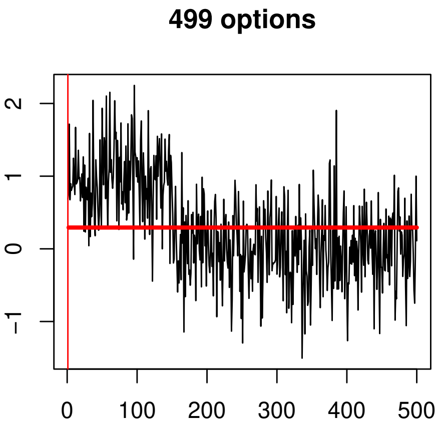
Online vs Offline



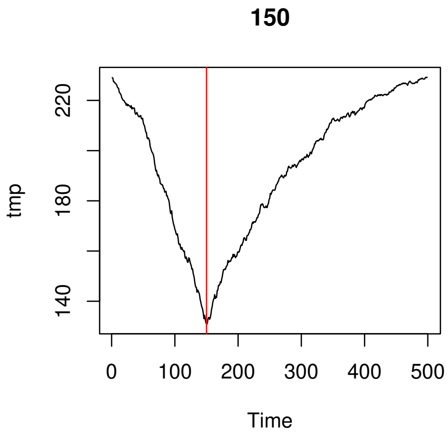
Fitting changepoint models



Finding a single change



Finding a single change



Finding a single change

How do we know if the changepoint found is “significant” or not?

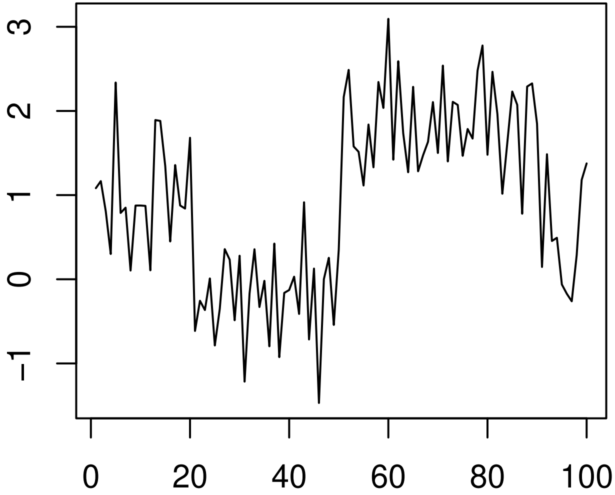
- We also calculate the cost of the whole data with no change
- If the difference is **large enough** then we say there is a change

In practice **large enough** is hard to define as it is application dependent.

The default in the `changepoint` package is MBIC - a Modified Bayesian Information Criterion.

This will not be appropriate for all data sets! More later . . .

Multiple Changepoints



Multiple Changepoints

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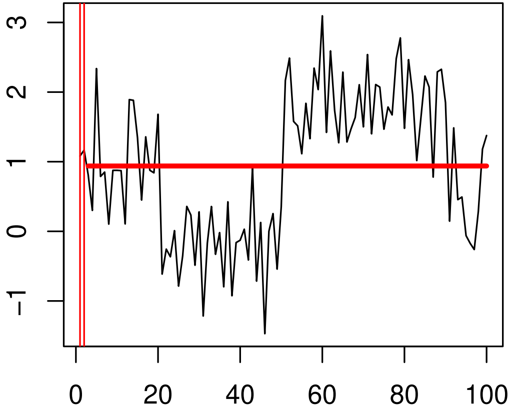
4851 options



The Challenge

- What are the values of $\tau_1, \ldots, \tau_m$?
- What is m?
- For n data points there are 2^{n-1} possible solutions
- If m is known there are still $\binom{n-1}{m}$ solutions
- If $n = 1000$ and $m = 10$, 2.607755×10^{23} solutions
- How do we search the solution space efficiently?

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PELT!

Teaser for later

We will:

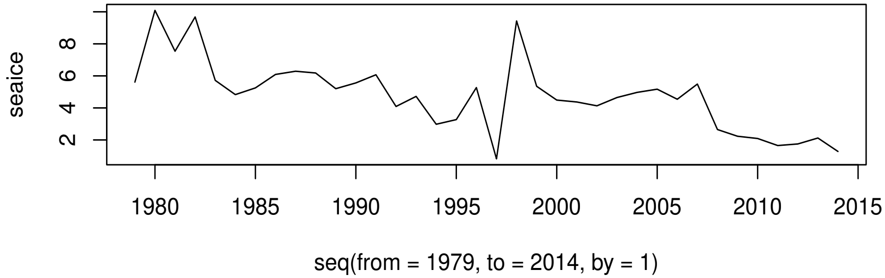
- learn how to use the `changepoint` package to fit standard changepoint models
- discuss practical aspects of fitting changepoint models including penalty choice
- learn how to use the `EnvCpt` package to automatically chose the best model from a set of models

All through a set of data examples from sea ice, air quality, soil moisture and surface/sea temperatures.

Sea Ice

Yearly Sea Ice measurements for July-Sept for Barents from 1979 until 2014.

```
plot(seq(from=1979,to=2014,by=1),seaice,type='l')
```



Air Quality

Air quality data, daily O_3 normalised measurements at several sites in London from 2020. Use the “TH4” coded site.

```
load('AQ_data_KCL_2020.RData')
plot(df_2020_out[df_2020_out$code=="TH4",c(1,5)],type='l')
```